

ROYAL FITNESS CLUB

Nutrition & Healthy Eating Certificate

MODULE 8

Final Assessment & Certificate

Duration	20 minutes
Course Code	CS_B2
Level	Beginner
Format	Study Guide PDF

Part of the Royal Fitness Club Professional Certification Program.

ASSESSMENT INSTRUCTIONS

How to Complete This Assessment

- 25 questions covering all 7 modules of CS_B2 Nutrition & Healthy Eating
- One correct answer per question (A, B, C, or D)
- Passing score: 70% (18/25). Open-book assessment — refer to module notes
- Complete in order; target 20 minutes maximum

1. Leucine activates muscle protein synthesis primarily through which intracellular pathway?

- A) AMPK activation → fatty acid oxidation
- B) mTORC1 activation → ribosomal biogenesis and MPS
- C) FOXO3 → autophagy induction
- D) NF-κB → pro-inflammatory cytokine release

2. Which food pairing creates a complete essential amino acid profile from two incomplete plant proteins?

- A) Broccoli + cauliflower
- B) Almonds + walnuts
- C) Rice (or roti) + lentils (dal)
- D) Spinach + kale

3. Resistant starch is formed when cooked starchy foods are:

- A) Eaten immediately after cooking while hot
- B) Frozen and then reheated rapidly
- C) Cooled after cooking (retrograde crystallisation)
- D) Processed into flour

4. The omega-6:omega-3 ratio in modern diets is approximately:

- A) 1:1 (balanced)
- B) 4:1 (ancestral)
- C) 15:1 to 20:1 (pro-inflammatory excess)
- D) 2:1 (slight imbalance)

5. Which Vitamin D3 blood level is considered optimal for athletes?

- A) <20 ng/mL
- B) 20–29 ng/mL
- C) 40–60 ng/mL
- D) >100 ng/mL

6. What is the recommended strategy to enhance non-haem iron absorption?

- A) Consume with calcium-rich dairy foods
- B) Drink tea with the meal
- C) Consume with vitamin C-rich foods (lemon, amla, bell pepper)
- D) Eat with foods high in phytates (wheat bran)

7. Curcumin bioavailability is most effectively enhanced by:

- A) Taking on an empty stomach with water
- B) Combining with black pepper (piperine) and dietary fat
- C) High-dose supplementation (5000mg+)
- D) Converting to a synthetic form

8. The highest Vitamin C content per gram among common foods is found in:

- A) Orange
- B) Lemon
- C) Kiwi fruit
- D) Amla (Indian gooseberry)

9. KSM-66 refers to:

- A) A synthetic anabolic compound in sports nutrition
- B) A standardised ashwagandha root extract with established clinical evidence
- C) A creatine formulation with magnesium
- D) A type of whey protein hydrolysate

10. In a caloric surplus with high insulin levels, which scenario is correct?

- A) High insulin always causes fat gain regardless of caloric balance
- B) Fat gain occurs only when in a caloric surplus — insulin drives storage in that context
- C) Carbohydrate intake causes fat gain independent of total calorie intake
- D) Insulin prevents muscle protein synthesis at elevated levels

11. Total metabolic adaptation after significant weight loss (10%+ body weight) can reduce TDEE by approximately:

- A) 50–100 kcal/day
- B) 100–200 kcal/day
- C) 300–500 kcal/day
- D) 700–1000 kcal/day

12. Which carbohydrate absorption transporter uses fructose as its substrate?

- A) SGLT1
- B) GLUT4
- C) GLUT5
- D) GLUT2

13. The maximum glucose absorption rate in the small intestine from a single carbohydrate source (e.g., maltodextrin alone) is approximately:

- A) 30g/hour
- B) 60g/hour
- C) 90g/hour
- D) 120g/hour

14. Pre-workout nutrition should MINIMISE which two macronutrients to reduce GI distress?

- A) Protein and carbohydrates
- B) Carbohydrates and water
- C) Fat and fibre
- D) Sodium and potassium

15. Which supplement has the strongest peer-reviewed evidence base for improving strength and power?

- A) BCAAs
- B) Testosterone boosters
- C) Creatine monohydrate
- D) L-carnitine

16. Caffeine's primary mechanism as an ergogenic aid is:

- A) Increasing testosterone production acutely
- B) Directly stimulating muscle contraction force
- C) Blocking adenosine receptors and reducing perceived fatigue
- D) Increasing glycogen breakdown rate

17. Whey ISOLATE differs from whey CONCENTRATE primarily in:

- A) Higher fat content in isolate
- B) Higher protein content (90%+) and lower lactose in isolate
- C) Faster absorption rate of concentrate
- D) Different amino acid profile

18. Beta-alanine works primarily for efforts lasting:

- A) <10 seconds (sprint)
- B) 1–4 minutes (glycolytic)
- C) >60 minutes (endurance)
- D) All durations equally

19. Which magnesium form has the highest bioavailability and is best suited for sleep support?

- A) Magnesium oxide
- B) Magnesium sulphate (Epsom salt)

- C) Magnesium glycinate
- D) Magnesium carbonate

20. The "diet break" strategy recommends eating at maintenance calories for 1–2 weeks to:

- A) Build muscle mass during a fat loss phase
- B) Restore leptin, reduce ghrelin, and counteract metabolic adaptation
- C) Allow liver glycogen to deplete completely before restarting
- D) Test whether the previous deficit was too aggressive

21. Antinutrients in legumes (dal) are best reduced by:

- A) Eating raw legumes with vitamin C
- B) Grinding into flour and consuming at high temperature
- C) Soaking 8–12h + pressure cooking + sprouting or fermentation
- D) Adding salt to cooking water only

22. An athlete weighing 70kg performing a 90-minute high-intensity session in moderate heat should consume approximately:

- A) Water only — no carbohydrates or electrolytes needed
- B) 30–45g carbohydrates per hour + sodium 200–500mg/hour
- C) 90g carbohydrates per hour + no sodium
- D) Only BCAAs intra-workout

23. For a vegan athlete, which supplement is NON-NEGOTIABLE (must supplement; no reliable food source)?

- A) Vitamin C
- B) Magnesium
- C) Vitamin B12
- D) Potassium

24. SARMs (Selective Androgen Receptor Modulators) should be avoided because:

- A) They are legal but slightly less effective than creatine
- B) They are unapproved drugs with documented liver toxicity, hormonal suppression, and are banned in sport
- C) They only work for women; not men
- D) They require a doctor prescription but are otherwise safe

25. The Dietary Inflammatory Index (DII) is improved most by increasing which food category?

- A) Processed meats and refined grains
- B) Omega-6 seed oils
- C) Fatty fish, colourful vegetables, and polyphenol-rich foods
- D) High-GI carbohydrates and saturated fat

ANSWER KEY

Q1: B	Q2: C	Q3: C	Q4: C	Q5: C
Q6: C	Q7: B	Q8: D	Q9: B	Q10: B
Q11: C	Q12: C	Q13: B	Q14: C	Q15: C
Q16: C	Q17: B	Q18: B	Q19: C	Q20: B
Q21: C	Q22: B	Q23: C	Q24: B	Q25: C

ROYAL FITNESS CLUB NUTRITION & HEALTHY EATING CERTIFICATE

This certifies that

—
has completed the

Nutrition & Healthy Eating Certificate — CS_B2

demonstrating comprehensive knowledge of macronutrients, micronutrients, Indian superfoods, meal planning, metabolism, sports nutrition, and evidence-based supplementation.

Instructor Signature

Date